

ECO491 PSET HW6

Written Solutions

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Fall 2025

Last updated: December 12, 2025

Question 1: Currency Option Risk

1.1 Traditional Historical Simulation

We consider a European call option on the USD/BRL exchange rate with maturity $T = 0.25$ years. Under the Garman–Kohlhagen framework, the option value depends on the spot exchange rate S_t , the domestic interest rate $r_{d,t}$, the foreign interest rate $r_{f,t}$, and the implied volatility σ_t .

A traditional Historical Simulation (HS) for this option is implemented as follows. First, we identify the relevant risk factors $(S_t, r_{d,t}, r_{f,t}, \sigma_t)$ and compute their historical daily changes $(\Delta \log S_t, \Delta r_{d,t}, \Delta r_{f,t}, \Delta \sigma_t)$. Each historical vector of changes defines a scenario.

For each scenario, the historical changes are applied to today's observed risk-factor levels:

$$S^{(t)} = S_0 e^{\Delta \log S_t}, \quad r_d^{(t)} = r_{d,0} + \Delta r_{d,t}, \quad r_f^{(t)} = r_{f,0} + \Delta r_{f,t}, \quad \sigma^{(t)} = \sigma_0 + \Delta \sigma_t.$$

The option is then re-priced under each scenario using the Garman–Kohlhagen formula. This produces a distribution of hypothetical option prices and, by comparison with today's price, a distribution of profit-and-loss (P&L) outcomes.

Because the option is fully re-priced under each historical shock, this approach constitutes a *full-valuation* Historical Simulation rather than a delta-normal approximation. No parametric assumptions are imposed on the joint distribution of risk factors.

Figures below plot the risk factors used to construct the HS scenarios.

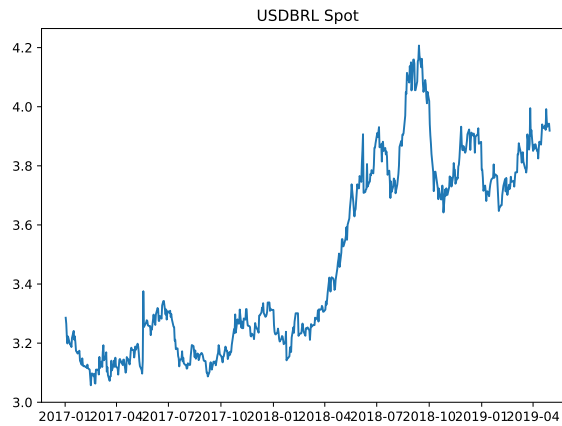


Figure 1.0.1: USD/BRL spot exchange rate.

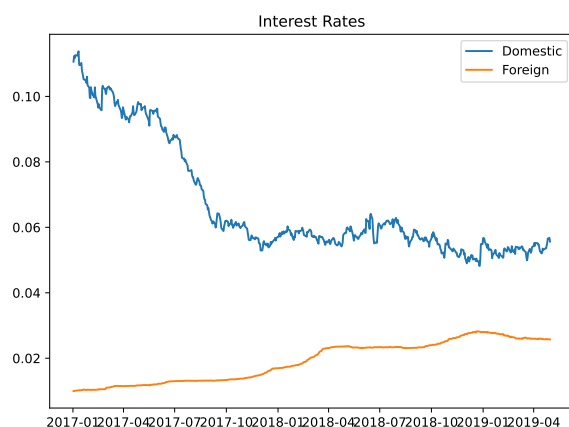


Figure 1.0.2: Domestic and foreign interest rates.

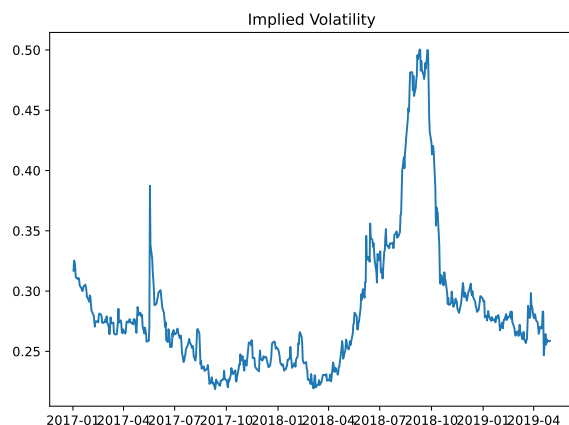


Figure 1.0.3: Implied volatility.

1.2 HS Value at Risk

Using the full available historical dataset as a single estimation window, the option is re-priced under each historical scenario, yielding an empirical distribution of P&L. The 1% Historical Simulation Value at Risk is defined as the negative 1% quantile of this distribution:

$$\text{VaR}_{1\%} = -q_{0.01}(\text{P\&L}).$$

Figure 4 shows the resulting HS P&L distribution. This measure captures the option's nonlinear exposure to the exchange rate, interest rates, and volatility, as well as their empirical dependence structure.

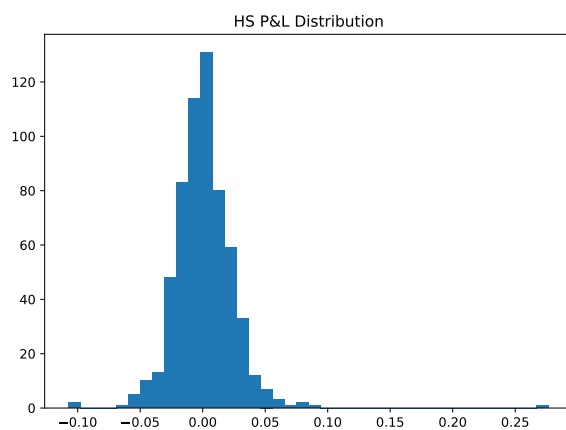


Figure 1.0.4: Full-valuation HS P&L distribution for the currency option.

Question 2: Backtesting Value at Risk

2.1 HS and RiskMetrics VaR: graphical analysis

We backtest two Value at Risk models for Apple daily log returns over the period 2015–2021: a 250-day Historical Simulation (HS) model and a RiskMetrics EWMA model with decay parameter $\lambda = 0.94$. Both models target a 1% tail probability.

At each date t , the VaR forecast is constructed using the previous 250 returns and compared to the realized return R_t . A violation (or hit) occurs when

$$R_t < -\text{VaR}_t.$$

Figures 5 and 6 plot realized returns together with the HS and RiskMetrics VaR forecasts. Violations are marked at their true realized return values, allowing direct assessment of both the frequency and severity of tail exceedances.

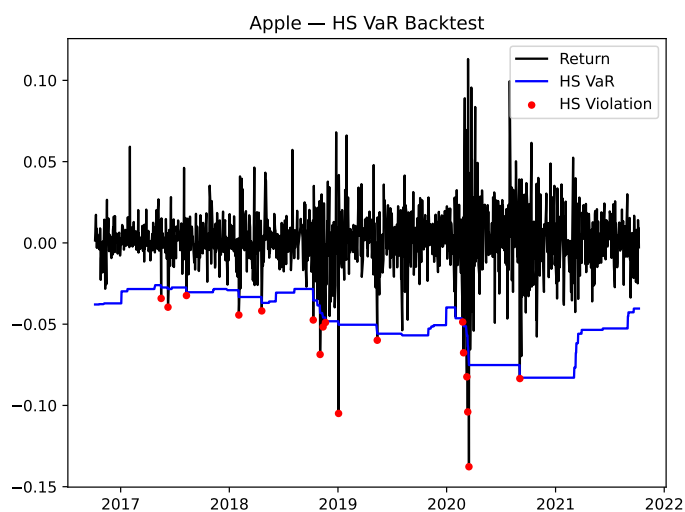


Figure 2.0.1: Apple HS VaR backtest with violations overlaid.

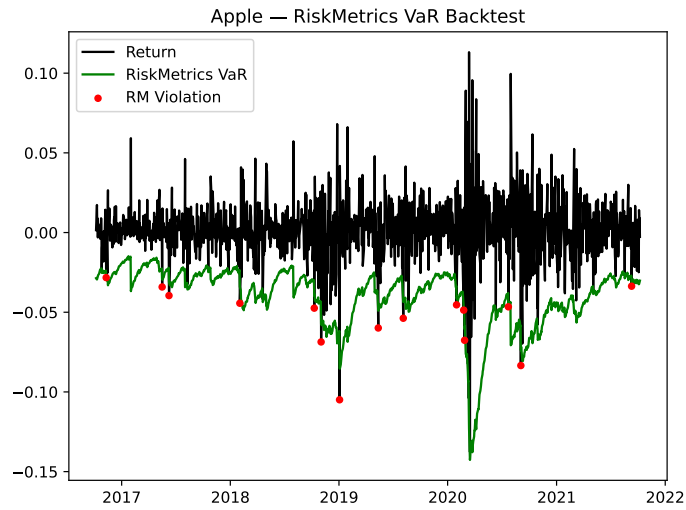


Figure 2.0.2: Apple RiskMetrics VaR backtest with violations overlaid.

To further highlight the timing and magnitude of tail events, Figures 7 and 8 isolate the violations for each model. Each violation is plotted at its actual realized return, preserving the scale of losses.

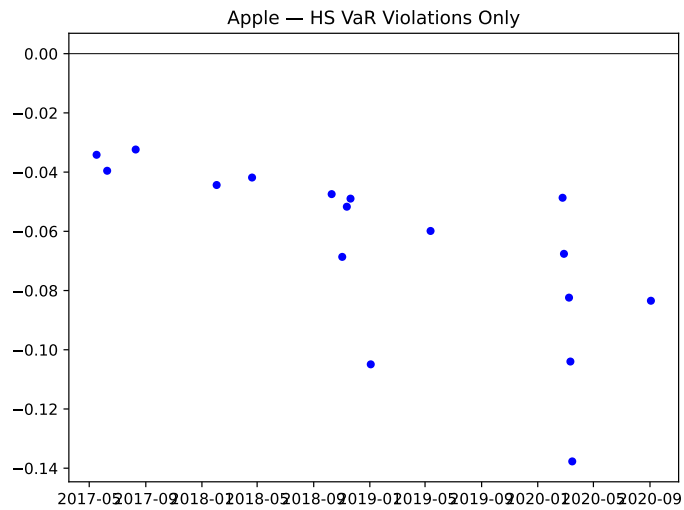


Figure 2.0.3: Apple HS VaR violations only (actual returns).

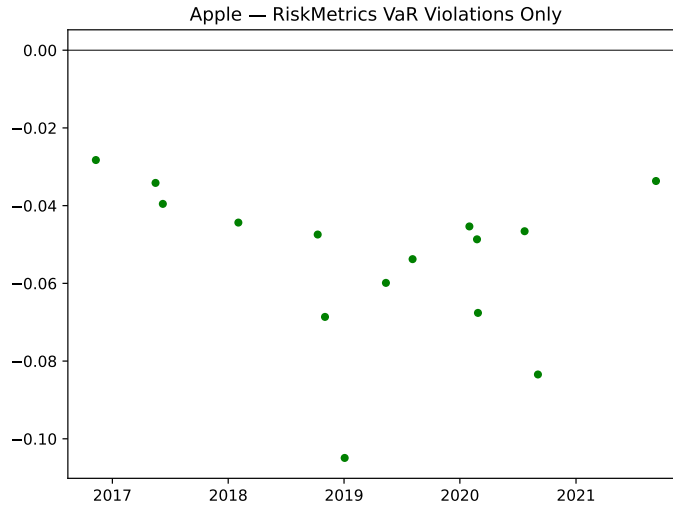


Figure 2.0.4: Apple RiskMetrics VaR violations only (actual returns).

Based on these graphs, we expect approximately 1% of observations to be violations under correct calibration. If the observed violation frequency is close to this level and violations do not exhibit excessive clustering, we would expect the formal backtesting procedure below not to reject the model. Systematic over- or under-frequency of violations would instead suggest rejection.

2.2 Unconditional coverage test

We formally assess VaR calibration using the Kupiec unconditional coverage (UC) test. Let $I_t = \mathbf{1}\{R_t < -\text{VaR}_t\}$ denote the violation indicator. Under the null hypothesis of correct calibration, $\mathbb{P}(I_t = 1) = p$ with $p = 0.01$.

The UC likelihood ratio statistic is

$$LR_{uc} = -2 \log \left(\frac{(1-p)^{T-T_1} p^{T_1}}{(1-\hat{\pi})^{T-T_1} \hat{\pi}^{T_1}} \right), \quad \hat{\pi} = \frac{T_1}{T},$$

where T is the number of backtest observations and T_1 is the number of violations. Under the null, LR_{uc} is asymptotically χ_1^2 .

The test statistics computed for the HS and RiskMetrics models are compared to the 10% critical value of the χ_1^2 distribution to assess whether the observed violation frequencies are consistent with the nominal 1% level.